

ChatGPT

Kendall's Tau Overview

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Kendall's Tau

Kendall's Tau (τ)

Kendall's Tau is a statistical measure used to determine the **strength and direction of association between two rankings**.

It tells us:

- whether two variables/rankings increase together,
- decrease together,
- or are unrelated.

It is commonly used in:

- ranking systems,
- recommendation systems,
- social network analysis,
- information retrieval,
- comparing centrality rankings in graphs,
- machine learning evaluation.

Intuition

Suppose two judges rank the same students.

If both judges mostly agree on ordering:

- Kendall's Tau $\rightarrow$ close to **+1**

If they mostly disagree:

- Kendall's Tau $\rightarrow$ close to **-1**

If rankings are random:

- Kendall's Tau $\rightarrow$ around **0**

Core Idea: Pairwise Agreement

Instead of comparing exact values, Kendall's Tau compares **pairs of items**.

For every pair (i, j) :

- If both rankings order them the same way → **Concordant pair**
- If rankings disagree → **Discordant pair**

Mathematical Definition

Basic Formula

$$\tau = \frac{C-D}{\binom{n}{2}}$$

Where:

- C = number of concordant pairs
- D = number of discordant pairs
- $\binom{n}{2}$ = total number of pairs

Concordant vs Discordant

Suppose:

Item	Rank X	Rank Y
A	1	1
B	2	3
C	3	2

Consider pair (B,C):

- In X: $B < C$
- In Y: $B > C$

Order changes → **Discordant**

Step-by-Step Example

Suppose:

$$X = [1, 2, 3, 4]$$

$$Y = [1, 3, 2, 4]$$

Total pairs:

$$\binom{4}{2} = 6$$

Pairs:

Pair	X Order	Y Order	Type
(1,2)	same	same	Concordant
(1,3)	same	same	Concordant
(1,4)	same	same	Concordant
(2,3)	2<3	3>2	Discordant
(2,4)	same	same	Concordant
(3,4)	same	same	Concordant

So:

- $C = 5$
- $D = 1$

Therefore:

$$\tau = \frac{5-1}{6} = 0.667$$

Strong positive correlation.

Range of Kendall's Tau

τ Value	Meaning
+1	Perfect agreement
0	No association
-1	Perfect disagreement

Why Kendall's Tau is Useful

Unlike Pearson correlation:

- it does **not assume linearity**
- works on **rankings**
- robust to outliers
- captures **ordinal relationships**

Kendall Tau vs Pearson Correlation

Feature	Kendall Tau	Pearson
Uses ranks	Yes	No
Measures monotonic relation	Yes	No
Sensitive to outliers	Less	High

Feature	Kendall Tau	Pearson
Linear assumption	No	Yes

Kendall Tau vs Spearman Rank Correlation

Both use rankings, but:

Spearman

Uses difference in ranks.

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2-1)}$$

Kendall

Uses pairwise agreement/disagreement.

Kendall Tau is often considered:

- more interpretable,
- more statistically robust,
- better for small datasets.

Spearman is computationally faster.

Time Complexity

Naive computation:

$$O(n^2)$$

because all pairs are checked.

Optimized methods using merge sort:

- $O(n \log n)$

Variants of Kendall's Tau

Kendall Tau-a

Basic version.

Kendall Tau-b

Handles ties.

Kendall Tau-c

Used for rectangular tables.

Most libraries use **Tau-b**.

Handling Ties (Tau-b)

When equal ranks exist, denominator changes:

$$\tau_b = \frac{C-D}{\sqrt{(C+D+T_x)(C+D+T_y)}}$$

Where:

- T_x = ties in X
 - T_y = ties in Y
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Applications

In Machine Learning

- compare ranking models
- evaluate recommendation systems
- feature importance comparison

In Social Networks

You may use Kendall Tau to compare:

- PageRank ranking
- Betweenness ranking
- K-shell ranking
- influence rankings in multilayer networks

Example:

If two centrality algorithms produce similar node rankings, Kendall Tau will be high.

Python Example

```
Python
from scipy.stats import kendalltau

x = [1,2,3,4]
y = [1,3,2,4]

tau, p = kendalltau(x, y)

print(tau)
```

Output:

```
Python
0.6666666667
```

Geometric Interpretation

Kendall Tau measures:

“How frequently pairwise ordering is preserved.”

It is fundamentally a **pair-order consistency metric**.

Important Insight

Kendall Tau focuses on:

relative ordering

not actual numeric distance.

So:

- [1, 2, 3]
- [10, 20, 30]

have perfect Kendall correlation.

In Network Science Context

In your multilayer influence-ranking work, Kendall Tau is extremely important because you often compare:

- predicted influential nodes
- actual diffusion performance
- rankings from different centrality methods

Example:

Method A Rank	Method B Rank
Node1	Node2
Node2	Node1

Kendall Tau quantifies how similarly the methods rank nodes.

High Tau:

- methods agree.

Low Tau:

- methods behave differently.